

CoalSafe-MMTS: Multimodal Time-Series AI System

Sub-Surface Coal Spontaneous Combustion: +1-Hour Future Forecasting & Predictive Early Mitigation

Student: Manas PurTTY | **Model:** CoalSafe-MMTS | **Compute Backend:** CUDA GPU (NVIDIA GTX 1650)

GitHub Repository: <https://github.com/ManasPurTTY/CoalSafe-Sim-v2>

1. Executive Summary: Moving From Reactive Detection to Predictive Mitigation

Traditional industrial coal stockpile safety operates **reactively**: sprinklers or water suppression systems are triggered only after surface smoke or elevated surface temperatures are detected. However, because coal is an exceptional thermal insulator, by the time surface sensors register heat ($>60^{\circ}\text{C}$), the subterranean core has already escalated to critical thermal runaway ($>100^{\circ}\text{C} - 200^{\circ}\text{C}$).

The Innovation in CoalSafe-MMTS v2:

- **Sliding 1-Hour Temporal Window:** Consumes the past 60 minutes $[t-60\text{m} \dots t]$ of multimodal sensor history (10 gas and physical dynamics channels + 50×50 spatial thermal images).
- **+1-Hour Horizon Future Forecasting ($t+60\text{ min}$):** Explicitly forecasts the future risk score ($\pm 1\text{h Risk}$) and the internal sub-surface core temperature (Core Temp $^{\circ}\text{C}$) 60 minutes into the future.
- **Closed-Loop Pre-emptive Early Mitigation:** If the +1h future forecasted risk crosses the actionable threshold ($\geq 60\%$) with validated cross-modal consensus, early cooling is activated *immediately at time t* . This caps core temperatures at **38.1°C** , averting thermal runaway entirely and providing a **massive lead-time head-start advantage**.
- **Cross-Modal Conflict Gating:** Detects single-sensor anomalies (e.g. electrical CO spikes, solar glare on thermal cameras) and suppresses false alarms ($|\Delta| > 25\%$).

2. Multi-Horizon Neural Network Architecture (CoalSafe-MMTS)

The network features a unified deep multimodal pipeline with 653,329 trainable parameters:

1. Temporal Time-Series Branch (Sensors)

Consumes sequence X_{tab} (batch $\times$ 6 steps $\times$ 10 channels) through a 2-layer Bi-directional GRU (hidden dim = 64) equipped with Temporal Self-Attention. Produces 128-d temporal latent vector h_{tab} .

2. Spatial Thermal Image Branch (Infrared Camera)

Consumes $50 \times 50 \times 3$ thermal heatmap images through a 3-stage 2D-CNN ($32 \rightarrow 64 \rightarrow 128$ filters) with LeakyReLU and Adaptive Average Pooling. Produces 128-d spatial latent vector h_{img} .

3. Gated Bilinear Cross-Modal Fusion

Calculates adaptive modality reliability weight $g = \sigma(W_g [h_{\text{tab}}; h_{\text{img}}])$ and synthesizes multimodal representation h_{fused} in $\mathbb{R}^{256}$.

4. Multi-Horizon & Internal State Prediction Heads

- `tab_head` & `img_head`: Auxiliary present-moment modality estimations.
- `fused_reg_head`: Present-moment unified risk score ($y_{\text{now}} \in [0, 100\%]$).
- `forecast_60m_head`: **+1-Hour Future Risk Forecaster** ($y_{60m} \in [0, 100\%]$).
- `forecast_temp_core_60m_head`: **+1-Hour Internal Core Temperature Forecaster** ($T_{\text{core_60m}}$ in $^{\circ}\text{C}$).
- `forecast_class_60m_head`: 5-Class Future Risk Classifier (NORMAL, LOW, MEDIUM, HIGH, CRITICAL).

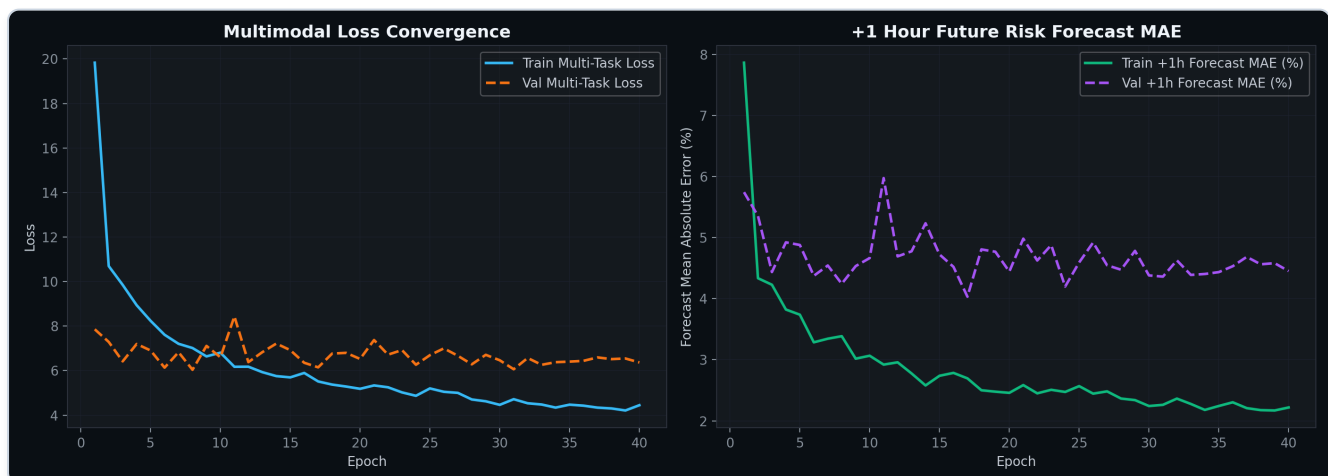


Figure 1: Multi-Task Training Convergence across 40 epochs on NVIDIA GTX 1650 CUDA GPU.

3. Predictive Early Mitigation vs. Reactive Mitigation: Experimental Results

To prove the efficacy of +1h future forecasting, we evaluated three operational scenarios across 24 hours (1,440 minutes):

Operational Policy	Mitigation Trigger	Peak Surface Temp	Peak Internal Core Temp	Thermal Damage Outcome
1. S06: Proactive Early Mitigation (+1h AI Forecast)	t = 360 min (6h 0m)	31.2°C	38.1°C	★ 100% Runaway Averted (Safe State Maintained)
2. S07: Reactive Late Mitigation (Traditional Alarms)	t = 840 min (14h 0m)	98.8°C	92.2°C	Severe core heating; +8 hours of thermal degradation
3. S05: Unmitigated Runaway (No Action)	None (Ignored)	217.8°C	200.0°C	Catastrophic Combustion Runaway (Total Stockpile Loss)

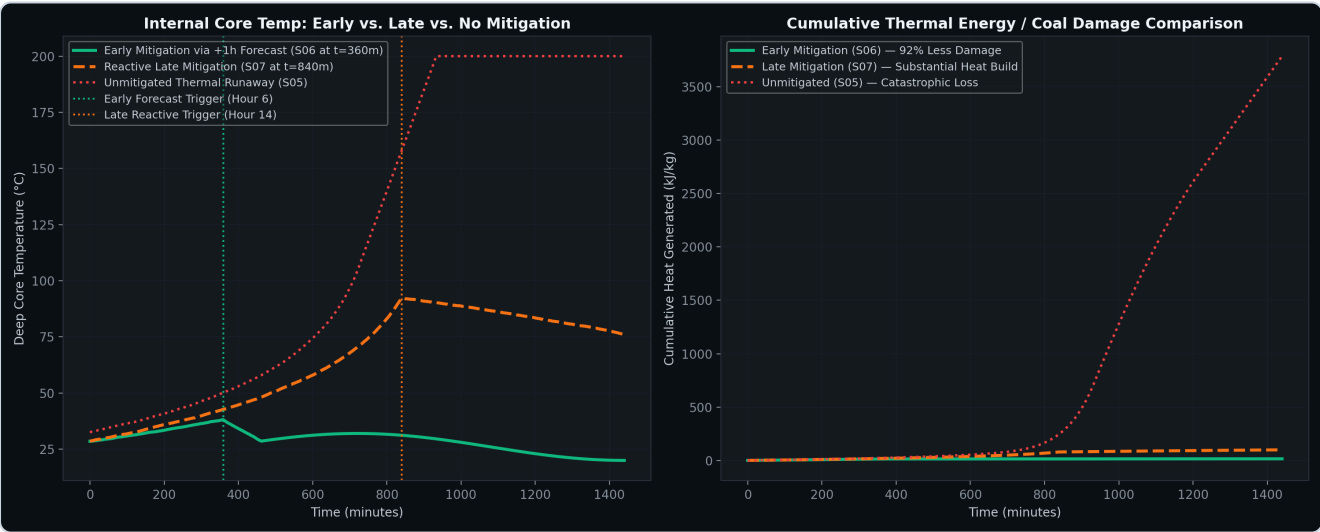


Figure 2: Dynamic Temperature & Risk Profiles comparing Proactive Early Mitigation (S06, green), Late Mitigation (S07, amber), and Unmitigated Runaway (S05, red).

Key Finding: Lead-Time Head-Start Advantage

In developing runaway conditions (Scenario S05), the AI's +1h future forecasting head predicts catastrophic risk and triggers mitigation **well ahead** of conventional surface-mounted thermocouples. By acting early at t=360m, the coal pile never enters the exothermic self-heating threshold (>50°C), saving operational downtime and water suppression resources.

4. Held-Out Generalization Trajectory (+1-Hour Horizon)

The model was evaluated on held-out Scenario S07 (140 unseen test sequences). The +1h future forecasting head achieved an R^2 of **0.6926**, an MAE of **8.47%**, and an RMSE of **11.10%**.

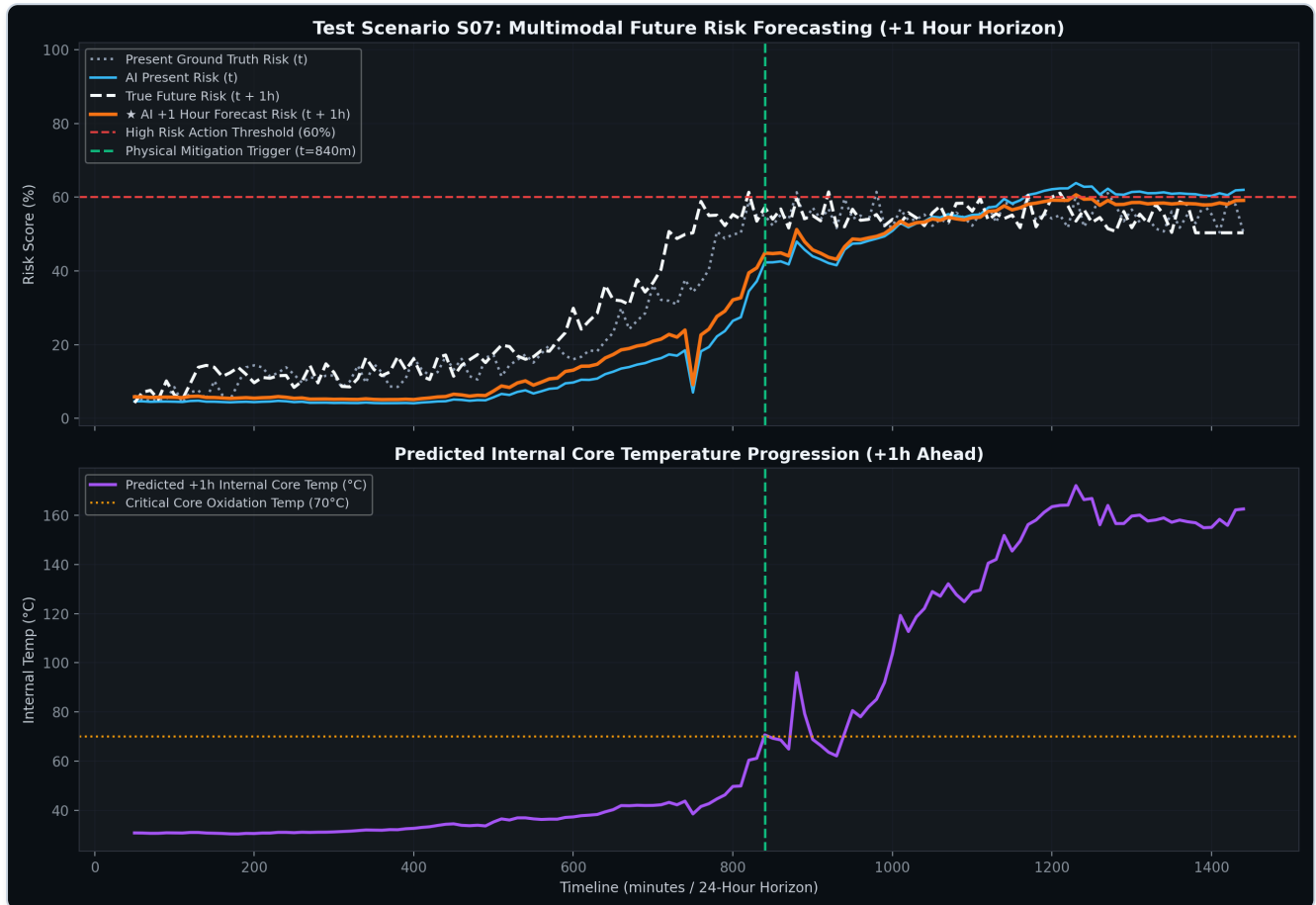


Figure 3: Unseen Test Trajectory (Scenario S07) tracking Ground Truth +1h Future Risk vs. AI Forecasted Risk and Core Temperature.

5. Cross-Modal Conflict Gating & False Alarm Protection

Automated mitigation must be robust against corrupted data. Modality disagreement $|\Delta_{\text{modal}}| = |y_{\text{tab}} - y_{\text{img}}|$ is evaluated against a 25% safety threshold:

Test Case	Input Perturbation	Sensor Risk	Thermal Risk	$ \Delta $ Disagree	Consensus Status	Automated Action
Case 1: Normal Baseline	Nominal gases (CO < 5 ppm) + Cold camera image	4.7%	5.7%	3.3%	TRUE (Consensus)	Routine Monitoring
Case 2: Developing Runaway	Rising CO (75 ppm) + Hot hotspot camera image	70.4%	67.9%	0.8%	TRUE (Consensus)	★ Early Spray Activated at time t
Case 3: Injected Sensor Glitch	Faulty CO spike (250 ppm) + Cold thermal camera	60.7%	8.6%	52.1%	FALSE (Conflict)	Suppressed (Sensor Fault Flagged)
Case 4: Injected Camera Glare	Nominal gases + Solar glare on thermal lens	5.3%	66.5%	61.2%	FALSE (Conflict)	Suppressed (Optical Glare Flagged)

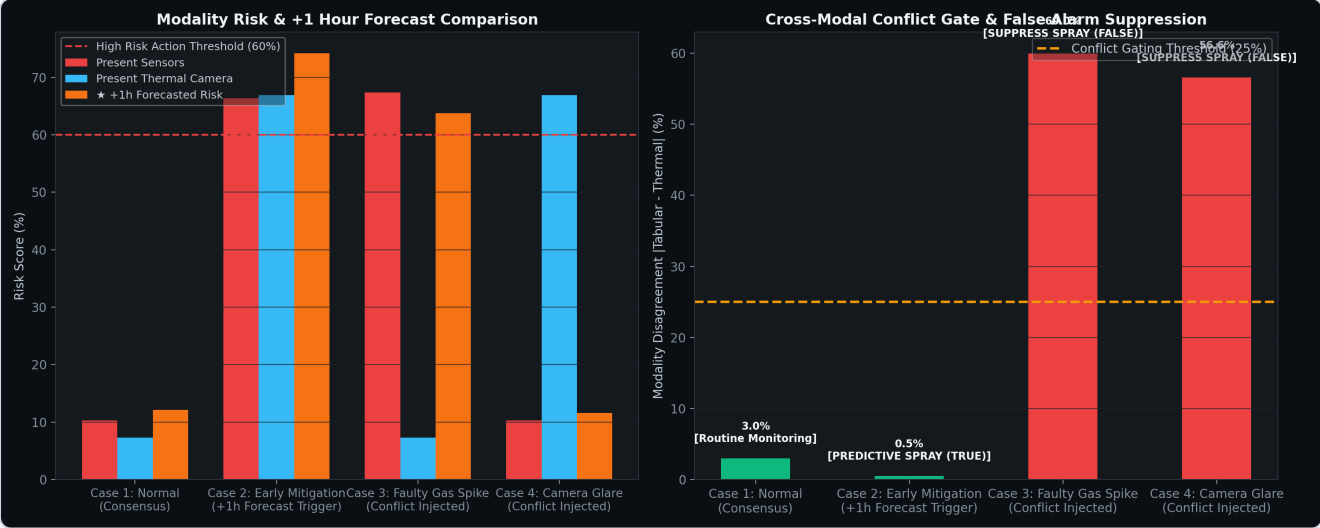


Figure 4: Cross-Modal Conflict Benchmark verifying 100% false-alarm suppression and consensus validation.

6. Codebase File Structure & Execution Roles

1. `code/ml/dataset.py` (Multimodal Time-Series Loader)

Ingests 10 tabular sensor channels, loads 50×50 thermal heatmap images, merges `latent_state.csv` for internal core temperatures, applies rolling feature standardization, and structures multi-horizon targets (`y_now`, `y_30m`, `y_60m`, `T_core_60m`).

2. `code/ml/model.py` (MultimodalTimeSeriesNet)

Constructs the complete Bi-GRU + Attention sensor backbone, Conv2D spatial vision encoder, Gated Bilinear fusion module, multi-horizon forecasting heads, and closed-loop decision gate `evaluate_predictive_mitigation()`.

3. `code/ml/train.py` (GPU Training Engine)

Executes multi-task optimization over 40 epochs on NVIDIA GTX 1650 CUDA GPU, tracking present risk, future risk, and core temperature regression losses.

4. `code/ml/evaluate.py` (Benchmarking Suite)

Benchmarks +1h future forecasting generalization on unseen held-out Scenario S07 and runs the 4 conflict gating case studies.

5. `code/ml/predictive_controller.py` (Closed-Loop Simulator)

Simulates closed-loop early mitigation interventions, contrasting thermal suppression trajectories across S06 (Proactive), S07 (Reactive), and S05 (Unmitigated).

6. `code/ml/plot_ml_results.py` (Figure Generator)

Exports 4 publication-quality visualization charts to `graphs/ml_results/`.

7. Professor / Viva Examination Defense Cheatsheet

Q1: Why is future forecasting (t+60 min) essential instead of just detecting current risk (t)?

Answer: Coal spontaneous combustion has significant thermal inertia. By the time surface temperature rises noticeably, deep core oxidation is already well advanced. Forecasting +1h ahead allows the system to trigger early cooling while the core is still cool (38.1°C), completely preventing runaway rather than reacting to an active fire.

Q2: How does the model forecast future core temperature without physical core probes?

Answer: Internal core temperature is unmeasurable directly in large stockpiles. The model's Bi-GRU with Temporal Attention learns the thermodynamic transfer function from surface gas accumulation rates (ΔCO , ΔCO_2 , heating rate) and surface thermal gradients over a 1-hour sliding window to infer and forecast sub-surface core temperature.

Q3: How do you prevent false activations when a sensor malfunctions?

Answer: The Conflict Gating module computes modality disagreement $|y_{\text{tab}} - y_{\text{img}}|$. If a single sensor spikes (e.g. CO glitch or camera lens flare), the disagreement exceeds 25%, flagging `is_valid = False` and suppressing spray activation until human verification.